

BRAC UNIVERSITY
CSE331 : Automata and Computability
Summer 2026

Duration: 30 minutes

Quiz 1

Total: 20 marks

Name: _____ **ID:** _____ **0.5 Points**

Section: _____ **0.5 Points**

DFA & RE

$L_1 \rightarrow \{w \in \Sigma^* \mid w \text{ contains 1 in the second position}\}$, where $\Sigma = \{0, 1\}$

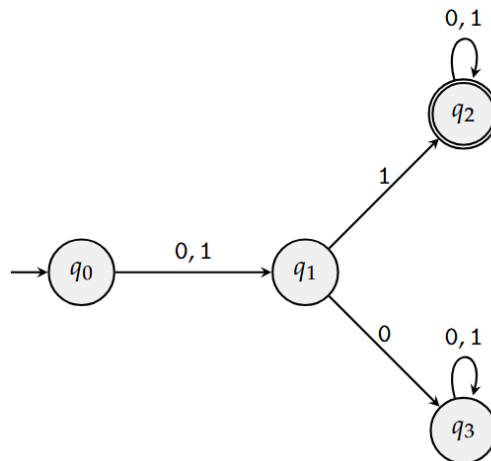
$L_2 \rightarrow \{w \in \Sigma^* \mid |w| \% 3 \neq 1\}$, where $\Sigma = \{0, 1\}$

$L_3 \rightarrow \{w \in \Sigma^* \mid \text{Every even position of } w \text{ is } 1\}$, where $\Sigma = \{0, 1\}$

$$L_4 = L_1 \cap \overline{L_2}$$

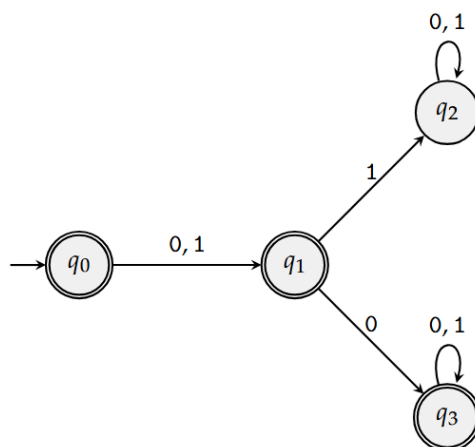
A. Draw the state diagram of a DFA for L_1 . (5 Points)

Answer:



B. Draw the state diagram of a DFA for $\overline{L_1}$. (2 Points)

Answer:



C. Write a Regular Expression for L_2 . (5 Points)

Answer: $(\Sigma\Sigma\Sigma)^*(\epsilon|\Sigma\Sigma)$

D. Write a Regular Expression for $(\overline{L_2})^*$. (1 Point)

Answer: $((\Sigma\Sigma\Sigma)^*\Sigma)^*$

E. Write all four letter strings in L_4 . (2 Points)

Answer: 0100, 0101, 0110, 0111, 1100, 1101, 1110, 1111.

F. Write a Regular Expression for L_4 . (3 Points)

Answer: $\Sigma 1 \Sigma (\Sigma \Sigma \Sigma)^* \Sigma$

G. Is $L_3 \subset L_1$? (1 Point)

Answer: No.